

Fundamentals of Computers and Computational Thinking

Course Code: CS101

Credits: 4 (Lecture: 3, Practical: 2)

Total Hours: 75 Hrs

MODULE II: Hardware Components & System Architecture

Comprehensive Academic Notes — Units 6 through 11 (Duration: 11 Hours, Weightage: 20 Marks)

Unit 6: Electronic Components — Active Components

CONCEPTUAL DEFINITION: ACTIVE ELECTRONIC COMPONENT

An **active electronic component** is a physical circuit element that relies on an external power source to operate, possesses the physical ability to dynamically control and modulate the flow of charge carriers (electrons and holes), and is capable of introducing electrical power gain, signal amplification, or high-speed non-linear digital switching into an electronic circuit.

The hardware of all modern computing systems is constructed from solid-state semiconductor electronics. Semiconductor materials, predominantly **Silicon (Si)** (atomic number 14, possessing four valence electrons in its outer shell) and occasionally **Germanium (Ge)** or **Gallium Arsenide (GaAs)**, form crystalline tetrahedral covalent lattice structures. In its pure (intrinsic) state, silicon is an insulator at absolute zero and a poor conductor at room temperature (**300 K**) because very few thermal electron-hole pairs exist in the conduction band.

Through the controlled introduction of minute chemical impurities — a process termed **doping** (typically 1 impurity atom per 10^6 to 10^8 silicon atoms) — the electrical conductivity of silicon can be increased by many orders of magnitude:

- **N-Type Semiconductor (Negative Extrinsic Silicon):** Formed by doping intrinsic silicon with pentavalent donor atoms (such as Phosphorus **P**, Arsenic **As**, or Antimony **Sb**, possessing 5 valence electrons). Four of the valence electrons form strong covalent bonds with adjacent silicon atoms in the crystal lattice, leaving the fifth valence electron loosely bound with an energy level just below the conduction band (**0.05 eV**). At room temperature, this donor electron is readily promoted to the conduction band. Consequently, **free mobile electrons act as majority charge carriers**, while thermally generated holes serve as minority carriers.
- **P-Type Semiconductor (Positive Extrinsic Silicon):** Formed by doping intrinsic silicon with trivalent acceptor atoms (such as Boron **B**, Aluminum **Al**, or Gallium **Ga**, possessing 3 valence electrons). The deficiency of one valence electron creates a localized vacancy or **hole** in the covalent bond structure. An electron from an adjacent bond can easily hop into this vacancy, causing the positive hole to propagate through the lattice. Consequently, **mobile positive holes act as majority charge carriers**, while free electrons serve as minority carriers.

FUNDAMENTAL ELECTRONIC CIRCUIT SYMBOLS (ACTIVE & PASSIVE)

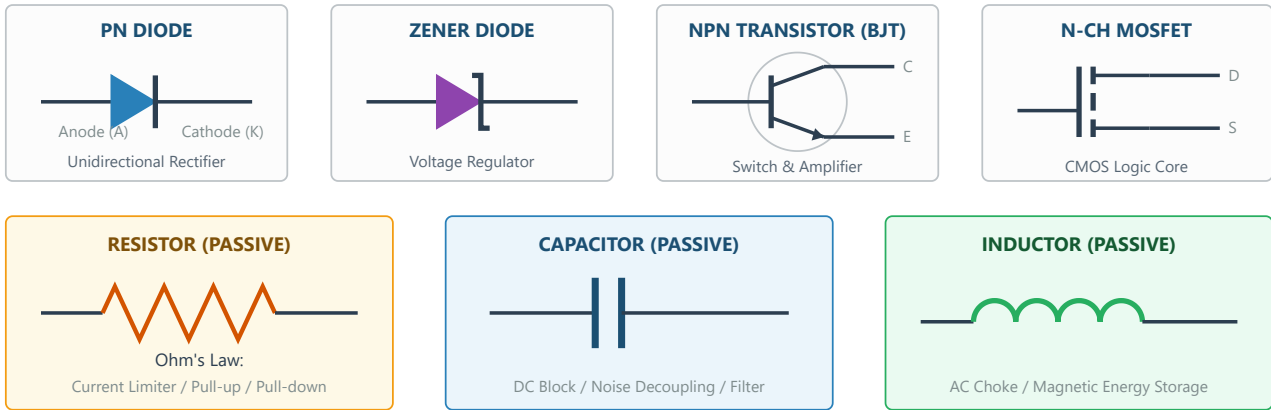


Figure 2.1: Active Semiconductor Components (Diodes, Transistors, ICs) and Passive Components (Resistors, Capacitors, Inductors).

6.1 Semiconductor Diodes

A **Diode** is a two-terminal, non-linear semiconductor device formed by metallurgically joining P-type and N-type semiconductor regions, creating an internal **PN Junction**.

1. Standard PN Junction Diode

- **Physical Operation & Barrier Potential:** Immediately upon PN junction formation, free electrons in the N-region diffuse across the junction to recombine with free holes in the P-region. This leaves behind fixed, uncompensated positive donor ions (P^+) on the N-side and fixed negative acceptor ions (B^-) on the P-side. This uncharged region containing fixed ions is termed the **depletion region**. The resulting internal electrostatic field creates a built-in barrier potential ($V_0 \approx 0.7 \text{ V}$ for Silicon at **300 K**, and $V_0 \approx 0.3 \text{ V}$ for Germanium).

- **Forward Bias:** When an external DC voltage source is connected with positive polarity to the P-region (Anode) and negative to the N-region (Cathode), the applied potential opposes the built-in barrier. When $V > 0.7 \text{ V}$ (the knee/threshold voltage), the depletion layer collapses, and current increases exponentially according to the Shockley Diode Equation:

$$I = I_s \left(e^{\frac{qV}{\eta kT}} - 1 \right)$$

where I_s is reverse saturation current, q is electron charge ($1.602 \times 10^{-19} \text{ C}$), k is Boltzmann constant ($1.38 \times 10^{-23} \text{ J/K}$), T is absolute temperature in Kelvin, and η is ideality factor ($1 \leq \eta \leq 2$).

- **Reverse Bias:** Connecting positive polarity to N and negative to P widens the depletion layer, preventing majority carrier flow. Only a minuscule, nanoampere-level reverse saturation current (I_s) flows due to thermally generated minority carriers, acting as an open switch.
- **Functions in Computing Systems:**
 - **AC to DC Rectification:** Converting **230V/110V** AC power from mains into DC voltage in power supply units (Half-Wave, Full-Wave Bridge Rectifiers).
 - **Transient Voltage Suppression (TVS):** Clamping electrostatic discharge (ESD) spikes on external USB and HDMI port lines to protect delicate microchips.
 - **Polarity Protection:** Preventing reverse-current damage to sensitive motherboard circuits.

2. Specialized Diode Types

- **Zener Diode:** A heavily doped PN junction engineered to operate reliably in the **reverse breakdown region** (Zener and Avalanche breakdown). When reverse voltage reaches the specified Zener breakdown threshold (V_Z), the voltage across the diode remains virtually constant despite wide changes in reverse current. *Function:* Precision voltage regulation and reference sources across motherboard VRMs.
- **Light Emitting Diode (LED):** Fabricated from direct bandgap compound semiconductors (such as Gallium Arsenide **GaAs** and Gallium Nitride **GaN**). Under forward bias, electrons and holes recombine directly across the bandgap, releasing discrete photons of light ($E = h\nu = \frac{hc}{\lambda}$). *Function:* Motherboard POST diagnostic debug LEDs, optical network transceivers, and drive activity indicators.
- **Photodiode:** A reverse-biased PN junction where incoming photons generate electron-hole pairs, producing a photocurrent proportional to light intensity. *Function:* Fiber-optic optical receivers and ambient light sensors.

6.2 Transistors: BJT and MOSFET

The **Transistor** is the foundational solid-state switching and amplifying element of all modern digital electronics.

1. Bipolar Junction Transistor (BJT)

A current-controlled, three-terminal semiconductor device consisting of two back-to-back PN junctions, constructed in either **NPN** or **PNP** configurations.

- **Terminals:**
 - **Emitter (E):** Heavily doped to inject maximum charge carriers into the base.
 - **Base (B):** Physically ultra-thin and lightly doped, controlling the transit of carriers.
 - **Collector (C):** Moderately doped and physically largest to dissipate thermal heat while collecting charge carriers.
- **Operational States:**
 - **Cut-Off State ($V_{BE} < 0.7V$):** Both emitter-base and collector-base junctions are reverse-biased. Collector current $I_C \approx 0$. Functions as an **Open Digital Switch (Logic 0)**.
 - **Saturation State:** Both junctions are forward-biased. Collector current reaches maximum value limited only by external circuit resistance ($V_{CE} \approx 0.2V$). Functions as a **Closed Digital Switch (Logic 1)**.
 - **Active Region:** Emitter-base forward-biased, collector-base reverse-biased. $I_C = \beta \cdot I_B$, where β is current amplification factor (**50–300**). Functions as an analog amplifier.

2. Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)

A voltage-controlled, four-terminal field-effect device (Gate, Drain, Source, Body/Substrate) where the metallic/polysilicon gate is electrically isolated from the semiconductor substrate by an ultra-thin dielectric insulating layer of Silicon Dioxide (**SiO₂**) or High- κ dielectric material (Hafnium Oxide **HfO₂**).

- **N-Channel Enhancement MOSFET (NMOS):** Applying a positive voltage to the Gate exceeding threshold voltage ($V_{GS} > V_{th}$) creates an electrostatic field that attracts electrons to the surface of the P-substrate, forming an inverted conductive N-channel between Source and Drain. Drain current I_D flows freely. When $V_{GS} = 0$, the channel vanishes, cutting off current.
- **Complementary MOS (CMOS) Technology:** Modern microprocessors combine NMOS and PMOS transistors in symmetrical complementary networks. In CMOS digital logic gates (e.g., CMOS Inverter / NOT gate), when the input is high, the NMOS pull-down network conducts to ground while the PMOS pull-up network is OFF; when the input is low, the PMOS conducts to V_{CC} while the NMOS is OFF. Because no direct

path to ground exists in steady state, CMOS circuits consume near-zero static DC power, dissipating energy almost exclusively during dynamic switching transitions:

$$P_{dyn} = C \cdot V_{DD}^2 \cdot f$$

6.3 Integrated Circuits (ICs)

An **Integrated Circuit (IC)** is a complete electronic circuit comprising millions to billions of interconnected microscopic transistors, diodes, resistors, and capacitors fabricated directly on a single monolithic semiconductor silicon crystal substrate using advanced photolithography, ion implantation, chemical vapor deposition, and plasma etching.

- **Scales of Integration:**
 - **SSI (Small Scale Integration):** < 10 logic gates (1960s, basic logic gates 7400 series).
 - **MSI (Medium Scale Integration):** 10 to 100 gates (multiplexers, decoders, counters).
 - **LSI (Large Scale Integration):** 100 to 10,000 gates (early 8-bit microprocessors, Intel 8080).
 - **VLSI (Very Large Scale Integration):** 10,000 to 1,000,000 gates (16/32-bit CPUs, Intel 80386/80486).
 - **ULSI (Ultra Large Scale Integration):** > 1,000,000 to 100 billion+ transistors (modern CPUs, GPUs, NPUs).
- **IC Packaging:** Dual In-line Package (DIP), Quad Flat Package (QFP), Ball Grid Array (BGA), and Land Grid Array (LGA). Modern microchips utilize multi-die 2.5D/3D Chiplet packaging interconnected via silicon interposers (e.g., TSMC CoWoS, Intel EMIB).

Unit 7: Electronic Components — Passive Components

CONCEPTUAL DEFINITION: PASSIVE ELECTRONIC COMPONENT

A **passive electronic component** is a circuit element that does not require an external power supply to function, cannot introduce net electrical power amplification into a circuit, and operates by dissipating, storing, or releasing electrical energy based on fundamental physical and electromagnetic laws.

Passive components form the indispensable foundation of all digital computer hardware. Without passive components to filter high-frequency noise, limit current surges, terminate high-speed transmission lines, and stabilize power rails, active microprocessors and memory chips cannot operate reliably.

7.1 Resistors

A **Resistor** is a two-terminal passive component designed to introduce precise electrical resistance (R) into a circuit, opposing the flow of electric charge in accordance with **Ohm's Law**:

$$\mathbf{V = I \cdot R \implies I = \frac{V}{R} \implies R = \frac{V}{I}}$$

The resistance is determined by material resistivity (ρ), length (L), and cross-sectional area (A):

$$R = \rho \frac{L}{A}$$

Electrical power dissipated as thermal heat is given by **Joule's Law**: $P = I^2 R = \frac{V^2}{R} = V \cdot I$.

Resistor Types, Color Coding & Tolerance

- **Resistor Construction Types:**

- **Carbon Film & Metal Film Resistors:** Thin resistive film spiraled around a ceramic rod, providing high precision (1% tolerance) and low thermal noise.
- **Wirewound Resistors:** Nichrome wire coiled around a ceramic core, capable of handling high power dissipation (5W to 50W) in power supply bleeder circuits.
- **Surface Mount Device (SMD) Resistors:** Rectangular ceramic chips (e.g., 0402, 0603, 0805 packages) with three-digit or four-digit EIA coding soldered directly onto motherboard PCB traces.

- **Standard 4-Band Resistor Color Code:**

$$R = (10 \cdot \text{Band 1} + \text{Band 2}) \times 10^{\text{Multiplier}} \pm \text{Tolerance} \%$$

Colors: Black (0), Brown (1), Red (2), Orange (3), Yellow (4), Green (5), Blue (6), Violet (7), Gray (8), White (9).

Tolerance: Gold ($\pm 5\%$), Silver ($\pm 10\%$).

Circuit Laws & Functions in Computer Hardware:

- **Kirchhoff's Current Law (KCL):** The algebraic sum of currents entering any circuit node equals zero: $\sum I_{in} = \sum I_{out}$. Essential for analyzing current distribution across parallel motherboard power rails.
- **Kirchhoff's Voltage Law (KVL):** The algebraic sum of all potential differences around any closed loop in a circuit equals zero: $\sum V = 0$. Used in voltage regulator feedback loop calculations.
- **Current Limiting:** Protecting sensitive LEDs, optocouplers, and transistor base junctions from over-current destruction ($R = \frac{V_{CC} - V_D}{I_D}$).
- **Voltage Division:** Scaling down motherboard voltage rails using series resistors:

$$V_{out} = V_{in} \cdot \left(\frac{R_2}{R_1 + R_2} \right)$$

- **Pull-Up and Pull-Down Resistors:** In digital logic buses (I^2C , PCIe SMBus, GPIO), an unconnected input pin will "float" between logic 0 and 1 due to electromagnetic noise. A **pull-up resistor** (4.7 k Ω to 10 k Ω) connects the line to V_{CC} to establish a default high state (1), while a **pull-down resistor** connects the line to Ground (0).
- **Bus Termination:** Matching transmission line characteristic impedance (50 Ω) on high-speed DDR memory and PCIe traces to prevent signal reflections and jitter.

7.2 Capacitors

A **Capacitor** is a passive component capable of storing electrical potential energy in an electrostatic field formed between two conductive parallel plates separated by an insulating **dielectric** medium.

$$Q = C \cdot V \implies C = \frac{Q}{V} = \frac{\epsilon_r \epsilon_0 A}{d}$$

where ϵ_0 is the permittivity of free space (8.854×10^{-12} F/m), ϵ_r is the relative dielectric constant, A is plate area, and d is plate separation distance.

The energy stored in the electrostatic field is $E = \frac{1}{2} CV^2$.

In AC circuits, a capacitor presents **Capacitive Reactance** (X_C), measured in Ohms (Ω):

$$X_C = \frac{1}{2\pi fC}$$

At DC ($f = 0 \text{ Hz}$), $X_C = \infty$ (blocks DC current). At ultra-high frequencies ($f \rightarrow \infty$), $X_C \rightarrow 0$ (acts as a low-impedance short circuit to ground for AC noise).

The transient charging and discharging of a capacitor across a series resistor is governed by the exponential equation with time constant $\tau = R \cdot C$:

$$V(t) = V_{max} \left(1 - e^{-\frac{t}{RC}}\right)$$

Capacitor Technologies & Hardware Roles:

- **Multi-Layer Ceramic Capacitors (MLCC):** Miniature surface-mount capacitors with low Equivalent Series Resistance (ESR) and low inductance. Placed by the hundreds on the underside of CPU/GPU sockets for **High-Frequency Decoupling and Power Plane Bypassing**.
- **Solid Polymer Aluminum Electrolytic Capacitors:** High-capacitance ($100\mu\text{F}$ to $1500\mu\text{F}$) capacitors with conductive polymer electrolytes, engineered to filter low-frequency voltage ripple on motherboard VRMs and prevent bulges or electrolyte drying.
- **DRAM Memory Cells:** In dynamic RAM, every single binary bit is stored as a minute charge (**30 femtofarads**) in a microscopic 3D trench capacitor.

7.3 Inductors

An **Inductor** (choke/coil) is a passive component consisting of a conductive insulated wire coiled around a magnetic core (ferrite or powdered iron), storing energy in a dynamic **magnetic field**.

Governed by **Faraday's Law of Induction** and **Lenz's Law**, any change in current induces an opposing back-EMF:

$$V_L = L \cdot \frac{dI}{dt}$$

The inductive reactance is $X_L = 2\pi fL$. An inductor allows pure DC ($f = 0$) to pass with zero reactive impedance, while opposing and choking high-frequency AC ripples. The energy stored in the magnetic field is $E = \frac{1}{2}LI^2$.

Functions in Computer Hardware:

- **Voltage Regulator Modules (VRMs):** High-current ferrite-core power chokes work alongside high-side and low-side switching MOSFETs in multi-phase buck-converter topologies to step down **+12V** SMPS power to ultra-stable **+1.1V** CPU core voltage with minimal ripple.
 - **Electromagnetic Interference (EMI) Suppression:** Toroidal ferrite chokes on USB and power cables filter high-frequency common-mode electromagnetic noise.
 - **Ferrite Beads:** Small SMD passive lossy beads that act as high-frequency resistors to attenuate RF noise on analog audio and clock traces.
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Unit 8: Motherboard Components — Core Subsystems

The **Motherboard** (system board / mainboard) is the central multi-layer Printed Circuit Board (PCB) — typically containing 6 to 12 copper layers — that serves as the physical and electrical backbone of a computer system, interconnecting the processor, memory, storage, expansion cards, and peripheral interfaces.

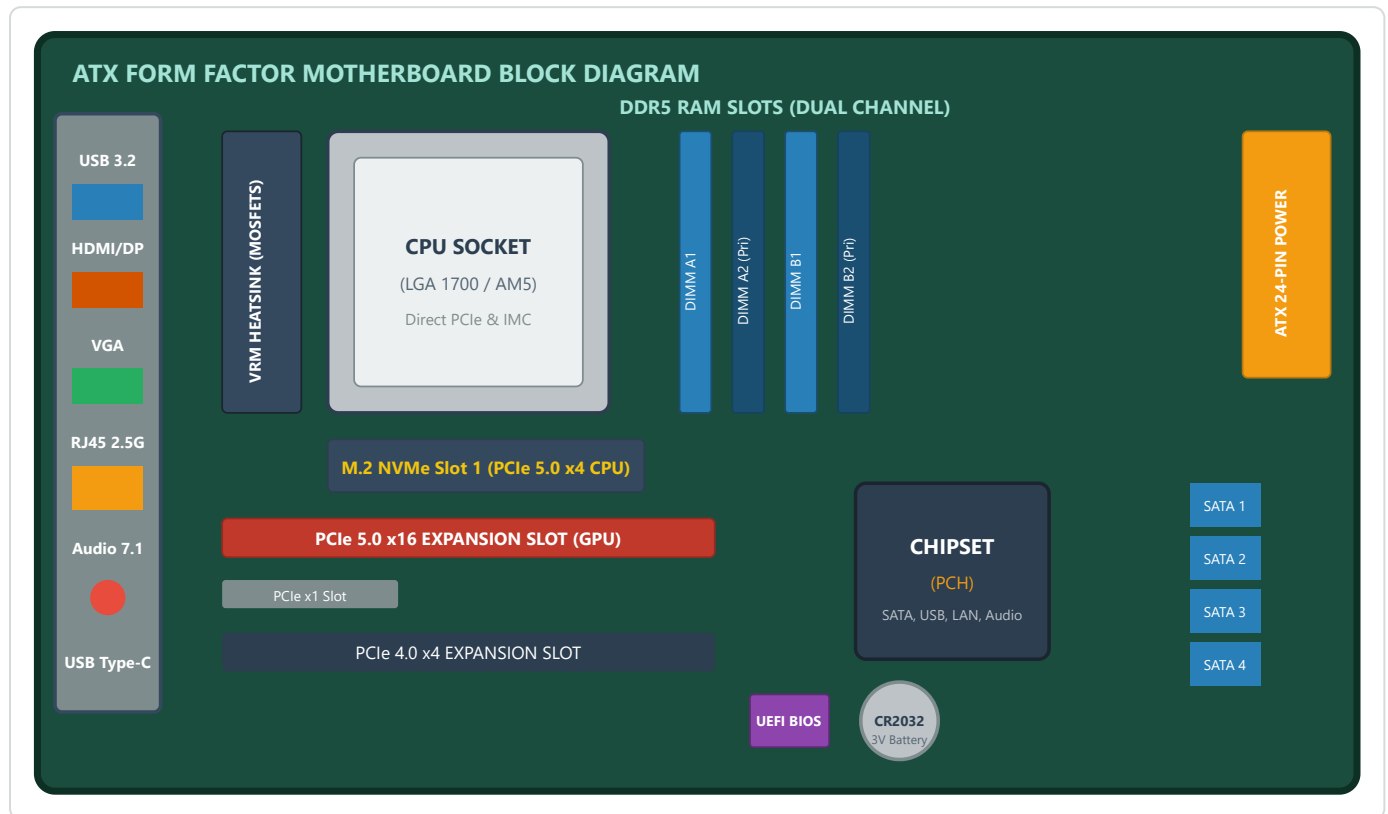


Figure 2.2: Comprehensive Architecture of Modern Motherboard Subsystems, Sockets, Slots, and Interconnects.

8.1 Form Factors & Mechanical Dimensions

- **Standard ATX (Advanced Technology eXtended):** 305 mm × 244 mm (12.0 in × 9.6 in). Up to 7 expansion slots, 4 DIMM slots, maximum VRM cooling phases.
- **Micro-ATX (mATX):** 244 mm × 244 mm (9.6 in × 9.6 in). Up to 4 expansion slots.
- **Mini-ITX:** 170 mm × 170 mm (6.7 in × 6.7 in). Compact form factor for small home-theater and portable PC builds, limited to 1 PCIe slot and 2 DIMM slots.

8.2 CPU Socket, VRM Power Stages & Cooling

- **Socket Architecture:**
 - **Land Grid Array (LGA):** Used primarily by Intel (e.g., LGA 1700, LGA 1851) and modern AMD (AM5). The delicate conductive pins are mounted inside the motherboard socket, while the underside of the CPU features flat gold contact pads. Reduces risk of damaging CPU pins during transit.
 - **Pin Grid Array (PGA):** Legacy architecture (e.g., AMD AM4) where rigid gold-plated pins project from the underside of the CPU package and insert into a Zero Insertion Force (ZIF) socket.
 - **Ball Grid Array (BGA):** Permanent surface-mount solder balls used in laptops, smartphones, and embedded SoC systems. Non-upgradable.

- **Voltage Regulator Module (VRM) Power Stages:** Modern CPUs consume **100A to 300A** at $\sim 1.1V$. Motherboards deploy multi-phase synchronous buck converters (e.g., 16+2+1 power phases using integrated Smart Power Stages / DrMOS). A PWM controller interleaves the phases with phase offsets ($\theta = \frac{360^\circ}{N}$), reducing current ripple and thermal stress.
- **Thermal Management & Cooling Solutions:**
 - **Thermal Design Power (TDP):** The maximum theoretical heat energy (in Watts) generated by the CPU under maximum real-world workload that the cooling system is engineered to dissipate.
 - **Thermal Interface Material (TIM):** Microscopic imperfections on the CPU Integrated Heat Spreader (IHS) and heatsink base trap air pockets (air is a poor thermal conductor: $k \approx 0.026 \text{ W/m} \cdot \text{K}$). Thermal paste (silicone suspended with zinc oxide, aluminum, or silver particles) fills these voids to maximize conductive heat transfer ($k \approx 5\text{--}12 \text{ W/m} \cdot \text{K}$).
 - **Active Air Cooling:** Copper heat pipes containing a low-boiling-point working fluid (evaporating at the CPU base, rising as vapor to aluminum fins, condensing, and returning via capillary wick action) paired with a 4-pin PWM (Pulse Width Modulation) controlled fan.
 - **All-In-One (AIO) Liquid Cooling:** Closed-loop system circulating liquid coolant via a mechanical water pump from a copper water-block across a multi-channel radiator cooled by static-pressure fans.

8.3 RAM Slots (DIMM) & Memory Channels

- **Form Factors:** Dual In-line Memory Modules (DIMMs, 288 pins on DDR4/DDR5) for desktops; Small Outline DIMMs (SO-DIMMs, 260/262 pins) for laptops.
- **Multi-Channel Architecture:**
 - **Single-Channel:** One 64-bit wide data bus between CPU memory controller and RAM.
 - **Dual-Channel:** Two independent 64-bit memory buses (**128-bit** aggregate width), doubling theoretical memory bandwidth. Requires population of matched pairs in alternating slots (typically slots A2 and B2).
 - **DDR5 Sub-channels:** DDR5 divides each DIMM into two independent 32-bit channels (plus 8-bit ECC), improving access efficiency.

8.4 Expansion Slots (PCI Express — PCIe)

PCIe is a high-speed, serial, point-to-point interconnect standard replacing older parallel PCI and AGP buses.

- **Differential Signaling Lanes:** Each PCIe lane (**x1**) consists of two pairs of differential copper traces (one pair for transmitting, one pair for receiving full-duplex). Slots are sized physically and electrically as **x1, x4, x8, and x16**.
- **PCIe Protocol Stack:**
 - *Transaction Layer:* Generates Transaction Layer Packets (TLP) for memory reads/writes and I/O.
 - *Data Link Layer:* Handles error detection (LCRC), ACK/NAK flow control, and retransmission.
 - *Physical Layer:* Handles 8b/10b (Gen 1/2) or 128b/130b (Gen 3/4/5) serialization and clock recovery.
- **PCIe Generations Bandwidth:**
 - **PCIe 3.0:** 8 GT/s (1 GB/s per lane simplex $\Rightarrow$ 16 GB/s on **x16**). Uses 128b/130b encoding (98.46% efficiency).
 - **PCIe 4.0:** 16 GT/s (2 GB/s per lane $\Rightarrow$ 32 GB/s on **x16**).
 - **PCIe 5.0:** 32 GT/s (4 GB/s per lane $\Rightarrow$ 64 GB/s on **x16**).

8.5 Chipset Architecture: Northbridge/Southbridge to Modern PCH

- **Legacy Architecture:**
 - **Northbridge (Memory Controller Hub - MCH):** Connected directly to CPU via Front Side Bus (FSB), handling high-speed subsystems: System RAM and AGP/PCIe graphics.
 - **Southbridge (I/O Controller Hub - ICH):** Connected to Northbridge, managing slower peripherals: PCI slots, SATA, IDE, USB, BIOS, and Audio.
 - **Modern Platform Controller Hub (PCH) Architecture:** Modern CPUs integrate the Memory Controller (IMC) and primary PCIe ~~x16~~ GPU and M.2 lanes directly on the CPU die. The motherboard contains a single **PCH chipset** connected to the CPU via a high-speed proprietary link (e.g., Intel DMI 4.0 or AMD PCIe uplink), managing secondary storage, USB controllers, SATA ports, Wi-Fi, and Gigabit Ethernet.
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Unit 9: Motherboard Components — Firmware, Storage & I/O Interfaces

9.1 BIOS / UEFI Firmware Subsystem

The **Firmware** is low-level non-volatile system software permanently stored on an onboard SPI Flash EEPROM chip on the motherboard.

- **Legacy BIOS (Basic Input/Output System):** Operates in 16-bit x86 Real Mode, constrained to 1MB of memory addressability. Relies on software interrupts (e.g., INT 13h) and the Master Boot Record (MBR).
- **Modern UEFI (Unified Extensible Firmware Interface):** 32-bit or 64-bit execution environment supporting rich graphical user interfaces, mouse input, direct network stack boot (PXE), and drive capacities exceeding **2.2 TB**.
 - **Secure Boot:** Cryptographic validation mechanism where the UEFI firmware verifies the digital signature of the OS bootloader against public keys embedded in NVRAM (Platform Key PK, Key Exchange Key KEK, Allowed Signature Database db, Forbidden Database dbx), protecting against low-level rootkits.
 - **ACPI (Advanced Configuration and Power Interface):** Power management standard defining processor power states (C0 active to C10 deep sleep) and system sleep states (S0 working, S3 suspend to RAM, S4 hibernation, S5 soft power off).
 - **CMOS Battery:** A lithium coin-cell battery (typically **CR2032, 3V**) powering the motherboard's Real-Time Clock (RTC) oscillator and maintaining non-volatile CMOS configuration registers when AC power is disconnected.

9.2 Storage Interfaces: SATA vs M.2 NVMe

- **Serial ATA (SATA III):** Operates at **6.0 Gbps (550 MB/s** net throughput) using the legacy **AHCI (Advanced Host Controller Interface)** protocol designed for spinning disks, limited to a single command queue of 32 commands with Native Command Queuing (NCQ).
- **M.2 NVMe (Non-Volatile Memory Express):** Compact form-factor drives (standard **22mm × 80mm**, form-factor 2280) connecting directly to PCIe lanes. Uses the lightweight **NVMe protocol**, supporting up to **64,000** parallel submission and completion queues each with **64,000** commands, unlocking speeds of **3,500 MB/s** (Gen 3), **7,500 MB/s** (Gen 4), and **14,000 + MB/s** (Gen 5).

9.3 External Display & Peripheral Ports

- **Video Interfaces:**
 - **VGA (Video Graphics Array):** 15-pin D-sub analog RGB interface. Subject to analog signal attenuation and noise at high resolutions.
 - **HDMI (High-Definition Multimedia Interface):** Digital audio/video interface using TMDS (Transition-Minimized Differential Signaling). HDMI 2.1 supports up to **48 Gbps** (4K @ 120Hz, 8K @ 60Hz, Dynamic HDR).
 - **DisplayPort (DP):** Digital packetized protocol similar to Ethernet; supports Daisy-Chaining via Multi-Stream Transport (MST).
 - **Universal Serial Bus (USB):**
 - **USB 2.0:** 480 Mbps (4-pin connector: V_{BUS} , $D-$, $D+$, GND).
 - **USB 3.2 Gen 1 (USB 3.0):** 5 Gbps (SuperSpeed, 9 pins).
 - **USB 3.2 Gen 2x2:** 20 Gbps dual-lane over Type-C.
 - **USB4 / Thunderbolt 4:** 40 Gbps, integrating PCIe tunneling, DisplayPort Alternate Mode, and USB Power Delivery (USB-PD up to 240W, 48V @ 5A). Reversible 24-pin Type-C connector.
 - **Network Interface (RJ-45 Ethernet):** 8-pin 8P8C modular connector interfacing onboard PHY transceiver with unshielded twisted pair (UTP Cat 5e/6/6a) cabling, supporting **1 Gbps**, **2.5 Gbps**, or **10 Gbps** full-duplex Ethernet using CSMA/CD and pulse-amplitude modulation.
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Unit 10: Computer Components — Power Supply & Storage Devices

10.1 Switched-Mode Power Supply (SMPS)

The **SMPS** is an electronic power supply unit that converts high-voltage alternating current (AC mains: 110V/230V, 50/60 Hz) into smooth, regulated low-voltage direct current (DC) rails required by internal computer hardware.

Detailed Operational Stages of an SMPS:

1. **Input EMI Filtering:** Inductive-capacitive network suppresses incoming line spikes and prevents high-frequency switching noise from feeding back into the electrical grid.
2. **Primary Rectification & Smoothing:** Bridge rectifier diodes convert AC to high-voltage DC, smoothed by large electrolytic reservoir capacitors ($\approx 320V$ DC).
3. **High-Frequency Inversion (Switching):** High-power MOSFET transistors switch the high-voltage DC ON and OFF at ultra-high frequencies (**50 kHz to 500 kHz**) using Pulse Width Modulation (PWM).
4. **Step-Down High-Frequency Transformer:** Because transformer core size is inversely proportional to operating frequency ($\text{Area} \propto \frac{1}{f}$), high frequency allows miniature, highly efficient ferrite transformers to step down voltages.
5. **Secondary Rectification & Filtering:** Schottky diodes rectify high-frequency pulses back to DC, smoothed by inductors and solid capacitors into clean output rails.
6. **Feedback & Supervisory Control:** Optocouplers provide galvanic isolation while monitoring output rails, adjusting PWM duty cycles dynamically, and asserting the **Power Good (PWR_OK / +5V)** signal to the CPU

only after voltages stabilize.

80 PLUS Energy Efficiency Certifications (at 50% Load)

Certification Tier	Efficiency at 20% Load	Efficiency at 50% Load	Efficiency at 100% Load
80 PLUS Standard	80%	80%	80%
80 PLUS Bronze	82%	85%	82%
80 PLUS Silver	85%	88%	85%
80 PLUS Gold	87%	90%	87%
80 PLUS Platinum	90%	92%	89%
80 PLUS Titanium	92%	94%	90%

Standard ATX Voltage Rails

Voltage Rail	Wire Color (ATX Standard)	Primary Hardware Consumers
+12V	Yellow	CPU VRMs, GPU PCIe power, cooling fans, spindle motors
+5V	Red	Motherboard logic circuits, SATA SSD/HDD logic, USB peripheral power
+3.3V	Orange	RAM slots, PCIe slot logic, Chipset, M.2 NVMe drives
+5VSB	Purple	Standby power (powers power button, Wake-on-LAN, RTC when PC is OFF)
PS_ON\#	Green	Active-low signal grounded by motherboard to turn on main DC rails

10.2 Storage Devices: HDD vs SATA SSD vs NVMe SSD

- **Hard Disk Drive (HDD):** Electromechanical magnetic storage. Data is recorded on spinning ferromagnetic aluminum/glass platters (**5400 or 7200 RPM**) divided into concentric *tracks*, *cylinders*, and *sectors* (**512e** or **4Kn** bytes). Read/write heads fly on an aerodynamic air bearing $\approx$ **10 nm** above platters on a voice-coil actuator arm.
 - **Performance Metrics:**

Average Access Time = Average Seek Time + Average Rotational Latency + Transfer Time

Mechanical Seek Time (**4–10 ms**) and Rotational Latency (**4.16 ms** at 7200 RPM) constrain random I/O performance to $\approx$ **75–150 IOPS**.

- **Recording Technologies:** Conventional Perpendicular Magnetic Recording (PMR/CMR) vs Shingled Magnetic Recording (SMR - overlapping tracks for high capacity at the expense of random write speed).
- **Solid State Drive (SSD):** Pure solid-state non-volatile semiconductor storage utilizing **NAND Flash Memory** (floating-gate or charge-trap transistors storing electrons).
 - **NAND Flash Hierarchy:** Memory is organized into *Dies* → *Planes* → *Blocks* (erase unit, **2–8 MB**) → *Pages* (read/write unit, **4–16 KB**). Crucially, NAND flash **cannot overwrite existing data in place**; a page must be erased (at the block level) before writing.
 - **NAND Cell Types:**
 - **SLC (Single-Level Cell):** 1 bit/cell (2 voltage levels). ~ 100,000 write cycles endurance, highest speed.
 - **MLC (Multi-Level Cell):** 2 bits/cell (4 voltage levels). ~ 10,000 cycles.
 - **TLC (Triple-Level Cell):** 3 bits/cell (8 voltage levels). ~ 3,000 cycles. Dominant consumer standard.
 - **QLC (Quad-Level Cell):** 4 bits/cell (16 voltage levels). ~ 1,000 cycles. High density, budget storage.
 - **Flash Management:** The SSD controller executes a **Flash Translation Layer (FTL)** handling *Wear Leveling* (distributing write/erase cycles evenly across blocks), **SLC Caching**, and **TRIM** command support (allowing OS to notify SSD when deleted file blocks can be proactively garbage-collected, preventing Write Amplification Factor — WAF degradation).

Unit 11: Memory Subsystem — RAM, ROM & Cache Hierarchy

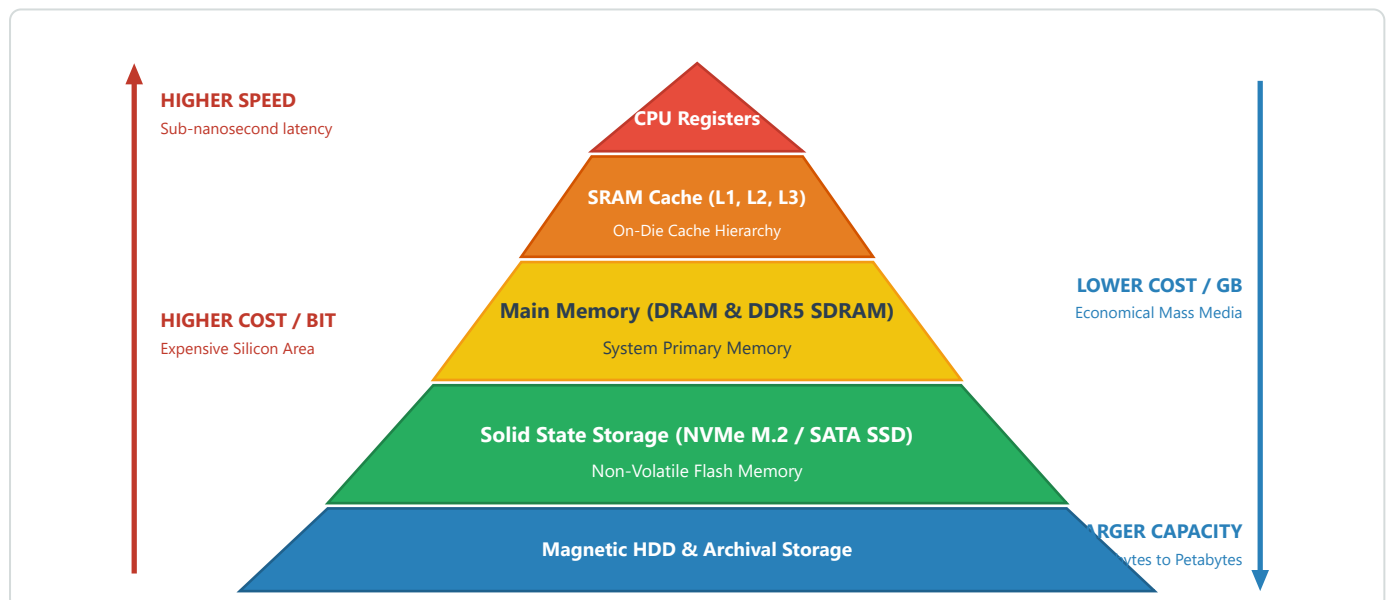


Figure 2.3: The Computer Storage and Memory Hierarchy displaying Latency, Capacity, and Cost Trade-offs.

11.1 Random Access Memory (RAM)

RAM is high-speed, volatile read-write semiconductor primary memory directly addressable by the CPU.

1. Dynamic RAM (DRAM) vs Static RAM (SRAM)

- **DRAM (1T-1C Cell):** Each bit is stored as an electrical charge on a single microscopic capacitor accessed via a single MOSFET pass-transistor. Because dielectric charge leaks over milliseconds, DRAM requires constant **periodic refresh cycles** (re-reading and rewriting every row every **64 ms**). High density and low cost make DRAM the standard for main memory.
- **SRAM (6T Bistable Latch):** Each bit is stored in a flip-flop circuit comprising six cross-coupled CMOS transistors. As long as power is supplied, SRAM holds its data indefinitely **without refresh cycles**. Access latencies are sub-nanosecond (**0.5–2 ns**), making SRAM the universal choice for CPU L1/L2/L3 on-die caches, though its larger physical size (**6×** silicon area of DRAM) makes it expensive.

2. Evolution of Synchronous Dynamic RAM (SDRAM & DDR)

SDRAM synchronizes all memory transfers to the system memory bus clock. **Double Data Rate (DDR)** transmits data on both the *rising edge* and *falling edge* of the clock signal, doubling theoretical throughput.

- **Prefetch Architecture:** DDR (**2n** prefetch), DDR2 (**4n**), DDR3 (**8n**), DDR4 (**8n**), DDR5 (**16n** prefetch).
- **DDR5 Key Advancements:** Integrated on-die Error-Correcting Code (ECC), dual independent 32-bit channels per module, on-module Power Management IC (PMIC) shifting voltage regulation from the motherboard to the DIMM, and operating voltage reduced to **1.1V**.
- **Memory Timings & CAS Latency:** Expressed as **CL-tRCD-tRP-tRAS**. **CAS Latency (CL)** is the exact clock cycle delay between the memory controller asserting a Column Address Strobe and data being placed on the bus.

$$\text{True Latency (ns)} = \frac{\text{CAS Latency} \times 2000}{\text{Data Rate (MT/s)}}$$

11.2 Read-Only Memory (ROM)

ROM is non-volatile primary memory that permanently retains instructions and data even when system power is completely disconnected.

- **Mask ROM:** Data permanently hardwired during factory semiconductor lithography. Completely unalterable.
- **PROM (Programmable ROM):** Manufactured blank with microscopic fusible links blown permanently using high electrical programming voltage. One-time programmable (OTP).
- **EPROM (Erasable PROM):** Uses floating-gate transistors. Erased by exposing the quartz glass window atop the IC package to intense Ultraviolet (UV) light for 20–30 minutes.
- **EEPROM (Electrically Erasable PROM):** Erased and reprogrammed byte-by-byte electrically using Fowler-Nordheim electron tunneling, without removal from the circuit board.
- **Flash Memory:** High-density block-erasable EEPROM. Modern UEFI BIOS chips use SPI Flash ROM.

11.3 Cache Memory Subsystem & Mapping Techniques

Cache Memory is ultra-fast SRAM placed between the CPU registers and main memory to bridge the processing-speed gap.

1. Principle of Locality of Reference

- **Temporal Locality:** If a memory location is accessed once, it is highly likely to be accessed again in the near future (e.g., loop counters, subroutine code).

- **Spatial Locality:** If a specific memory word is accessed, adjacent memory locations are highly likely to be accessed soon (e.g., sequential instruction execution, contiguous array traversal).

2. Cache Mapping Techniques & Address Partitioning

A physical memory address of m bits is partitioned into three fields: **Tag | Index / Set | Word Offset**.

1. **Direct Mapping:** Each main memory block maps to exactly one predetermined cache line:

$$\text{Cache Line} = (\text{Block Address}) \pmod{\text{Total Cache Lines}}$$

$$\text{Offset Bits} = \log_2(\text{Block Size}), \quad \text{Index Bits} = \log_2(\text{Number of Lines}), \quad \text{Tag Bits} = m - (\text{Index} + \text{Offset})$$

Advantage: Fast, simple tag comparison. *Disadvantage:* High conflict misses if two active memory blocks map to the same line.

2. **Fully Associative Mapping:** A main memory block can be loaded into *any* available cache line.

$$\text{Offset Bits} = \log_2(\text{Block Size}), \quad \text{Tag Bits} = m - \text{Offset}$$

Advantage: Zero conflict misses; maximum hit rate. *Disadvantage:* Requires expensive associative hardware comparators to search all cache tags in parallel.

3. **N -Way Set-Associative Mapping:** The cache is partitioned into S sets, each containing N cache lines ($S = \frac{\text{Total Lines}}{N}$). A memory block maps to a specific set based on index bits, but can occupy any of the N lines within that set.

$$\text{Set Index Bits} = \log_2(S), \quad \text{Tag Bits} = m - (\text{Set Index} + \text{Offset})$$

Represents the optimal engineering compromise in modern CPU L1/L2/L3 caches (e.g., 8-way or 16-way set associativity).

WORKED EXAMPLE 11.1: CACHE ADDRESS PARTITIONING

Problem: A 32-bit byte-addressable system has a **64 KB** 4-way set-associative cache with **64-byte** cache blocks. Calculate the bit-widths of the Word Offset, Set Index, and Tag fields.

Solution:

- Block Size = 64 bytes = 2^6 bytes $\implies$ **Offset Bits = 6 bits.**
- Total Cache Size = 64 KB = 65,536 bytes.
- Total Lines in Cache = $\frac{65,536}{64} = 1,024$ lines.
- Number of Sets $S = \frac{\text{Total Lines}}{N} = \frac{1024}{4} = 256$ sets = 2^8 sets $\implies$ **Set Index Bits = 8 bits.**
- Tag Bits = $32 - (\text{Set Index} + \text{Offset}) = 32 - (8 + 6) = 18$ bits.

The 32-bit address is partitioned as: **[Tag : 18 bits] | [Set : 8 bits] | [Offset : 6 bits]**.

3. Cache Write Policies & Replacement Algorithms

- **Write Policies:**
 - **Write-Through:** Every write to the cache is simultaneously written to main memory. High bus traffic, but memory is always completely synchronized.
 - **Write-Back:** Writes are made only to the cache line, marking a "Dirty Bit". The modified line is written back to main DRAM only when it is evicted by a replacement algorithm. Maximizes performance by

minimizing memory bus writes.

- **Cache Replacement Algorithms:**

- **Least Recently Used (LRU):** Evicts the cache block that has gone unreferenced for the longest duration (highest hit rate, requires hardware timestamp/counter tracking).
- **First-In First-Out (FIFO):** Evicts the oldest loaded block regardless of access recency.
- **Pseudo-LRU / Random:** Low-overhead approximations for high-associativity caches (**16**-way to **64**-way).